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Deep reinforcement learning-driven intelligent portfolio management with green computing: Sustainable portfolio optimization and management

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Highlights

- Proposes a dynamic predictor selection-based DRL model for adaptive trading strategies.
- Introduces a market environment evaluation module for dynamic portfolio optimization.
- Achieves superior Sharpe ratios using daily data from SSE 50 and CSI 500 indices.
- Enhances prediction accuracy and investment decisions via ablation experiments.
- Combines Green Computing principles with DRL for sustainable portfolio management.

Abstract

Portfolio management remains a key area in quantitative trading. To address limitations in existing deep reinforcement learning (DRL)-based models, which fail to adapt trading strategies and properly utilize supervisory information, we propose a Dynamic Predictor Selection-based Deep Reinforcement Learning (DPDRL) model. The DPDRL model integrates multiple predictors to forecast stock movements and dynamically selects the most accurate predictions, optimizing investment allocation via a market environment evaluation module. Our model was evaluated using daily candlestick data from the SSE 50 and CSI 500 indices. The results show that DPDRL outperforms other models in key evaluation metrics: it achieves a 48.99% Annualized Rate of Return (ARR), a Sharpe ratio of 2.34, an Annualized Volatility (AVoL) of 0.1390, and a Maximum Drawdown (MDD) of 8.21 %, significantly improving risk-return performance. Ablation experiments confirm the contributions of the dynamic predictor selector and market evaluation module to the model's accuracy and decision-making quality.

Introduction

In recent years, with the comprehensive deepening of reforms and the development of economic policies, China's market economy has grown rapidly, and the global economy has entered a new stage [1]. The international financial market is showing a trend of diversification and integration. In this context, the demand for wealth appreciation among enterprises and individuals has become increasingly urgent, manifested in the increasing variety and widespread attention of financial products. As people's understanding of financial management and investment deepens, they can choose from more financial products to achieve higher asset appreciation and returns. At present, this situation is becoming increasingly common, with stock and securities portfolio management being particularly common [2].

An investment portfolio is essentially a collection of multiple assets, which can include stocks, bonds, futures, options, and so on [3]. The process of constructing an investment portfolio refers to buying or selling various assets based on a certain proportion on a fixed principal basis, and continuously adjusting these proportions. By diversifying investments, it is possible to reduce holding risks and pursue the highest possible returns. In financial markets, the standard deviation of portfolio value is commonly used to measure risk, while the portfolio's profit margin is used to represent returns. For individual investors, optimizing asset allocation to achieve maximum long-term returns is very difficult. Even investment experts from financial institutions need to invest a lot of time and effort, applying professional knowledge to analyze each asset comprehensively in a noisy and unstable market. Therefore, traditional manual analysis is not only complex, but also costly [4].

With the continuous advancement of information technology, deep reinforcement learning (DRL) has shown great potential in the financial field [5]. The core idea of this method is to train and optimize the model through the interaction between the agent and the environment, which not only saves time in obtaining annotated samples, but also endows the model with the ability to learn autonomously. However, existing methods still face many unresolved issues. On the one hand, most methods only focus on a single prediction problem and fail to fully consider the cyclical characteristics of the stock market. Agents often only learn one market forecasting method, which may become unsuitable during rapid market changes. When there are significant differences in the

market, agents often need to learn new prediction methods at the cost of forgetting old ones, making it difficult to find suitable strategies in volatile situations. On the other hand, some investment strategies do not consider the proportion of funds, that is, they do not take into account the distribution ratio of the total investment amount among different stocks. Although deep learning can effectively predict the future trends of stocks and rank them reasonably based on market cyclical changes, it still lacks sufficient consideration in terms of the proportion of capital investment [6].

Considering the changes in investment strategies and economic cycles, this paper proposes a stock prediction method that takes into account different trading patterns to enhance the trading decision-making ability of the model. Agents learn appropriate trading strategies in different market environments at different times, improve the accuracy of predicting different assets, and optimize the allocation of investment funds. Most methods predict stock trends based on historical data, but different investment experts have different insights into the market. This article constructs an asset weight module to enable agents to learn different market interpretation methods to train multimodal predictors, while extracting time series features and combining different predictors to select the best prediction results in a reinforcement learning (RL) environment. In addition, considering the correlation between individual stock trends and market trends, this article also constructs a market environment evaluation module to dynamically adjust the proportion of funds. The main contributions of this article are as follows:

(1) In order to address the problem of single prediction, this paper designs a stock prediction method suitable for multiple trading models in RL networks. Agents will learn these trading patterns and choose the most suitable trading pattern based on the current market environment, thereby solving the problem of poor performance of agents in different time periods caused by a single prediction pattern.

(2) In order to solve the problem of capital ratio, this article divides the investment portfolio problem into two parts: investment weight problem and investment ratio problem, and designs two different modules to deal with these two problems, namely the asset weight module and the market environment evaluation module. The market environment evaluation module dynamically adjusts the capital ratio by analyzing market data to optimize the investment portfolio.

The remaining organization of this article is as follows: 2 provides a comprehensive review of related works, highlighting the key contributions and limitations of previous portfolio management models, including those based on DRL. 3 presents the methodology, where we introduce our DPDL model, explaining its components and how they address the limitations identified in existing models. 4 discusses the experimental setup and presents the results of our model, including comparisons with existing methods and performance metrics. Finally, 5 concludes the paper by summarizing the findings, discussing the implications of the results, and outlining directions for future work.

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Section snippets

Related works

Currently, research on investment portfolios remains a hot topic both domestically and internationally. This section mainly introduces the research status of traditional investment portfolio strategies, investment portfolio strategies based on time series prediction, and investment portfolio strategies based on DRL, and summarizes them

The traditional portfolio management analysis requires evaluating the returns and risks of each asset, but its results often lack credibility and stability due to ...

Problem description

The task of portfolio management is to maximize investment returns while assuming certain risks, and investment managers need to review the timing to change the weights of each asset class. For example, if an investment manager determines that the overall condition of a certain asset is more favorable to investors compared to other assets at the next moment, they need to transfer the weights of the investment portfolio to that asset. Here, the investment manager is our deep reinforcement ...

Experimental preparation

This section first introduces data processing, baselines, and measurement standards; Then evaluate the algorithm of this article using baselines in different markets; Finally, through ablation experiments, the impact of the dynamic selection predictor in the asset weight module and the market environment evaluation module on rewards was analyzed. The experiment is based on PyTorch 2 under Linux system 0.1 framework, hardware configuration is Intel Xeon CPU E5-2697R v4 @ 2.30 GHz, GPU RTX 3080. ...

Datasets

...

Conclusions

In this paper, we proposed the Dynamic Predictor Selection-based DPDRL model for portfolio management, which incorporates multiple predictors and a market environment evaluation module to improve prediction accuracy and optimize portfolio allocation. Our experiments on stock market data from the SSE 50 and CSI 500 indices demonstrate the effectiveness of our approach.

The DPDL model outperforms existing methods in key evaluation metrics, achieving an ARR of 48.99 %, an ASR of 2.34, an AVoL of ...

CRediT authorship contribution statement

Yi Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. ...

Ethical approval

This article does not contain any studies with human participants performed by any of the authors. ...

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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